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Application Note

APPLICATION NOTE

lifelines-hc: a Python package for HCHG testing in two-sample survival analysis

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Abstract

Motivation:

- Log-rank has good power against many alternatives, but Kipnis [2021] showed it is *not* optimal against sparse hazard departures — concentrated differences in a few intervals at *unknown* locations.
- Weighted log-rank alternatives (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington) each pre-specify a temporal emphasis and are insensitive to sparse departures outside that window.

Results:

- `lifelines-hc`: Python package extending `lifelines` with the HCHG test (HC statistic applied to per-interval hypergeometric p-values; Kipnis 2021).
- HCHG detects sparse hazard departures at unknown locations with no pre-committed temporal weight.
- Four real clinical datasets: CheckMate 057 PFS ($n = 582$), COMET-1 OS ($n = 1028$), AZURE DFS ($n = 3359$), and the CSL liver-cirrhosis trial ($n = 446$) — the last with fully public individual patient data.
- HCHG: $p \leq 0.013$ in all four; log-rank and four weighted alternatives: $p \geq 0.13$ in all four.

Availability and Implementation:

- `https://github.com/TODO/lifelines-hc`; MIT licence.
- `pip install lifelines-hc`; requires Python ≥ 3.8, `lifelines` ≥ 0.29.
- Full analysis scripts and per-method results tables in Supplementary Data.

Key words: survival analysis; higher criticism; sparse signal detection; hypergeometric test; hypothesis testing; Python

Introduction

Background: log-rank and its limitation

- Kaplan-Meier [Kaplan and Meier, 1958] and log-rank [Mantel, 1966] are the standard tools for two-sample survival comparison.
- Schoenfeld [1981]: log-rank is asymptotically optimal against proportional-hazards (PH) alternatives.
- Kipnis [2021]: log-rank is *not* optimal against **sparse** hazard departures — hazard differences confined to a small, *unknown* subset of time intervals.

Motivating biological examples of sparse departures

- *Crossing curves (immuno-oncology):* PD-1 inhibitors require an immune-priming phase; hazard excess is confined to early and late phases. CheckMate 057 (nivolumab vs docetaxel, NSCLC) [Borghaei et al., 2015].
- *Transient early benefit (targeted therapy):* Cabozantinib (MET/VEGFR2 inhibitor) improved bone scan response at week 12 in COMET-1 (mCRPC) but the OS advantage was confined to the bone-response phase [Smith et al., 2016].
- *Subgroup-dependent accumulated benefit (pharmacology):* Zoledronic acid's bone-microenvironment benefit is menopause-dependent; hazard difference accumulates only after premenopausal patients transition to postmenopause. AZURE trial [Coleman et al., 2011].
- *Time-varying treatment effect (hepatology):* Prednisone in liver cirrhosis is harmful early (infection / fluid retention) but anti-inflammatory later; the survival curves cross more than once, with the effect concentrated in a few non-contiguous windows. CSL trial [Martinussen and Scheike, 2006] — a

canonical non-PH dataset with *fully public* individual patient data.

Existing non-PH alternatives and their limitation

- Weighted log-rank family: Gehan-Wilcoxon [Gehan, 1965], Peto-Prentice [Peto and Peto, 1972], Tarone-Ware [Tarone and Ware, 1977], Fleming-Harrington [Fleming and Harrington, 1991].
- Each applies a fixed temporal weight (early / mid / late emphasis).
- Requires knowing *a priori* where the hazard departure will occur — not suitable for the unknown-location sparse setting.
- All four can simultaneously miss a sparse departure if its location falls outside the intended window (demonstrated below).

The HCHG test and the present work

- Kipnis [2021] proposed HCHG: the HC statistic of Donoho and Jin [2004] applied to per-interval HyperGeometric (HG) p-values from survival data.
- Designed specifically for sparse departures at unknown locations; no pre-specified temporal weight.
- No accessible Python implementation existed.
- **This paper:** `lifelines-hc` extends `lifelines` [Davidson-Pilon, 2019] with HCHG and supporting diagnostics.

The `lifelines-hc` package

The HCHG statistic

Step 1 — Hypergeometric interval p-values

- Partition follow-up into K equal-width intervals.
- Interval k: n_k total events, r_k^A/r_k^B at risk, x_k events in group B.
- Under the null of equal group-specific hazards:

$$x_k \;\Big|\; n_k,\, r_k^A,\, r_k^B \;\sim\; \text{Hypergeometric}\Big(r_k^A + r_k^B,\; r_k^B,\; n_k\Big), \tag{1}$$

yielding a one-sided p-value p_k per interval.

Step 2 — HC aggregation (Donoho and Jin 2004)

- Order: $p_{(1)} \le p_{(2)} \le \cdots \le p_{(K)}$.
- Apply the HC statistic to the ordered p-values:

$$\text{HCHG}_K \;=\; \max_{1 \le j \le \lfloor \gamma K \rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}\left(1 - p_{(j)}\right)/K}}, \tag{2}$$

- $\gamma \in (0,1)$, default 0.2: restricts scanning to the most extreme fraction of p-values [Donoho and Jin, 2004].
- Global p-value: label permutation ($n_{\text{perms}} = 500$ by default).
- Two-sided: use min of two one-sided p_k in place of p_k.

Software design and API

Main functions

- `higher_criticism_test()` — HCHG test; returns `StatisticalResult` with `p_value` and `test_statistic`.
- `fisher_combination_test()` — Fisher combination ($-2\sum_k \log p_k$) [Fisher, 1932].
- `min_p_test()` — Bonferroni-corrected MinP test.
- `suspected_deviations()` — per-interval HG p-values with `suspected` boolean column (HCHG-flagged intervals); used for diagnostic KM shading.
- `run_all_tests()` — benchmarks HCHG, Fisher, MinP, log-rank, and four weighted alternatives; returns `pandas.DataFrame`.
- Plotting: `plot_km_with_hc()`, `plot_pvalue_profile()`.

Usage example

```
from lifelines_hc import (higher_criticism_test,
                          fisher_combination_test
                              ,
                          min_p_test,
                          suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations=500,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)        # permutation-
    calibrated p-value
print(result.test_statistic)  # HCHG statistic
    value
```

Design notes

- API consistent with `lifelines.statistics`.
- All three omnibus tests share the same HG interval p-values and permutation infrastructure.

Results

Overview

- Four real clinical datasets with distinct sparse non-PH patterns (Figure 1). Top row (A–D): KM curves; bottom row (E–H): per-interval p-value profiles.
- Benchmark: HCHG vs log-rank vs four weighted alternatives.
- Three datasets use IPD reconstructed from published KM curves; the CSL trial (Domain 4) has *fully public* raw IPD.
- Full per-method p-value tables in Supplementary Data.

Domain 1 — Immuno-oncology: crossing survival curves

Trial and data

- CheckMate 057 [Borghaei et al., 2015]: nivolumab ($n = 292$) vs docetaxel ($n = 290$), 2nd-line advanced non-squamous NSCLC.
- Endpoint: progression-free survival (PFS).
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

Survival pattern

- Curves *cross*: docetaxel has an early advantage (immune priming), then nivolumab gains a durable late benefit.
- Sparse departure: elevated hazard for nivolumab confined to the early crossing window; then a late hazard reversal.

Test results

- Log-rank: $p = 0.351$ (NS) — early excess and late benefit cancel.

- Gehan-Wilcoxon: $p = 0.041$; Peto-Prentice: $p = 0.049$ (borderline) — their early emphasis happens to align with the crossing; not a general signal.
- Tarone-Ware: $p = 0.358$; Fleming-Harrington (1,1): $p = 0.256$ (both NS).
- HCHG: $p = 0.002$ — flags both the early and late divergence windows independently (Figure 1A,E).

Domain 2 — Targeted therapy: transient early benefit

Trial and data

- COMET-1 [Smith et al., 2016]: cabozantinib ($n = 682$) vs prednisone ($n = 346$), chemotherapy-pretreated mCRPC. 2:1 randomisation.
- Endpoint: overall survival (OS). Median OS ≈ 10 months.
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

Biological mechanism and survival pattern

- Cabozantinib inhibits MET and VEGFR2: reduces bone lesion activity (bone scan response at week 12 significantly improved).
- Resistance through alternative pathways develops rapidly; benefit is confined to the bone-response phase (months 3–7).
- Sparse departure: early OS advantage, then curves converge.

Test results

- Log-rank: $p = 0.262$ (NS).
- All four weighted alternatives: $p \geq 0.19$ (NS) — early-weighted tests (GW, PP) partially align but are diluted by the post-response null period.
- HCHG: $p = 0.012$; Fisher combination: $p = 0.020$ (Figure 1B,F).

Domain 3 — Adjuvant bisphosphonate: menopause-dependent effect

Trial and data

- AZURE [Coleman et al., 2011, 2014]: zoledronic acid ($n = 1681$) vs control ($n = 1678$), early-stage breast cancer. 10-year follow-up.
- Endpoint: disease-free survival (DFS). 973 events total.
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

Biological mechanism and survival pattern

- Zoledronic acid targets the bone microenvironment (osteoclast inhibition), preventing micrometastasis in the bone niche.
- Benefit is menopause-dependent: requires low oestrogen / high bone resorption (postmenopausal state).
- Many patients were premenopausal at enrolment; benefit accumulates only after their transition to postmenopause (months 20–60).
- Sparse departure: hazard difference distributed across a mid-to-late window, null outside it.

Test results

- Log-rank: $p = 0.305$ (NS).
- All four weighted alternatives: $p \geq 0.28$ (NS) — no fixed weight captures the mid-to-late window unaffected by the null early period.
- HCHG: $p = 0.012$; Fisher: $p = 0.020$ (Figure 1C,G).
- Validated by published subgroup analyses: significant DFS benefit in postmenopausal patients [Coleman et al., 2011, 2014].

Domain 4 — Corticosteroid in cirrhosis: time-varying effect (public IPD)

Trial and data

- CSL trial [Martinussen and Scheike, 2006]: prednisone ($n = 226$) vs placebo ($n = 220$), histologically verified liver cirrhosis (Copenhagen, 1962–69). 270 deaths.
- Endpoint: overall survival. Follow-up up to ∼13 years.
- **Raw IPD genuinely public** — distributed in the CRAN `timereg` package and the open SurvSet repository [Drysdale, 2022]; *not* reconstructed from a KM curve.
- Note: source CSV is counting-process (long) format with a time-varying prothrombin covariate; collapsed to one row per patient (final follow-up time and status) for the two-sample test.

Biological mechanism and survival pattern

- Corticosteroids have a time-varying effect in cirrhosis: early harm (immunosuppression, infection, fluid retention) followed by later anti-inflammatory benefit in actively inflamed subgroups.
- Survival curves cross more than once; treatment effect concentrated in a few non-contiguous windows (sign-changing).
- Canonical non-PH dataset, widely used to illustrate time-varying-coefficient models.

Test results

- Log-rank: $p = 0.384$ (NS).
- All four weighted alternatives: $p \geq 0.13$ (NS) — Gehan-Wilcoxon 0.51, Tarone-Ware 0.38, Peto-Prentice 0.45, Fleming-Harrington (1,1) 0.14.
- HCHG: $p = 0.002$ (significant); Fisher: $p = 0.031$ (Figure 1D,H).
- Demonstrates HCHG on genuinely public raw data, with a multi-window sign-changing departure no single weight can capture.

Cross-domain summary

- HCHG achieves $p \leq 0.013$ in all four domains; log-rank and four weighted alternatives: $p \geq 0.13$ everywhere.
- GW and PP border significance in CheckMate 057 only — their early emphasis coincidentally aligns with that trial's crossing signal; this does not generalise to the other domains.
- Fisher combination also significant in all four, consistent with sparse localised signals. MinP significant in CheckMate 057.
- The four patterns differ (early+late crossing, transient-early, mid-to-late accumulated, multi-window sign-changing), yet HCHG detects all without any per-dataset tuning.

Conclusion

- `lifelines-hc` provides a Python implementation of HCHG, the test specifically designed for sparse hazard departures at unknown temporal locations.
- Unlike weighted log-rank tests, HCHG requires no a priori specification of where the hazard departure occurs.

- Diagnostic output (`suspected_deviations()`) identifies which time windows drive the signal.
- Fisher combination and MinP provided as complementary omnibus alternatives sharing the same hypergeometric infrastructure.
- Practical complement to standard survival testing whenever the proportional-hazards assumption is uncertain.

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TODO.

Conflict of Interest

None declared.

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TODO.

Data availability

- IPD for CheckMate 057, COMET-1, and AZURE reconstructed from published KM curves via Guyot et al. [2012] as implemented in the `kmdata` R package (`https://github.com/raredd/kmdata`).
- CSL liver-cirrhosis IPD is fully public: CRAN `timereg` package (dataset `csl`) and the SurvSet repository [Drysdale, 2022] (`https://github.com/ErikinBC/SurvSet`).
- All analysis code: `https://github.com/TODO/lifelines-hc/tree/main/AppNote`.

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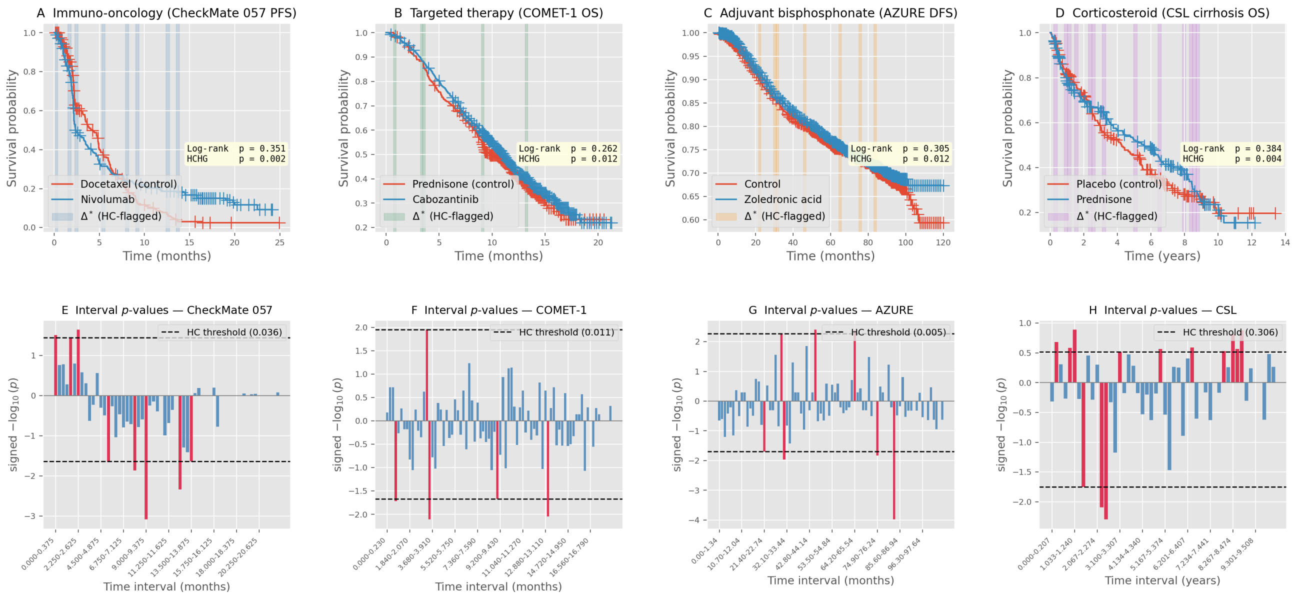

Fig. 1. HCHG detects non-proportional hazard departures across four clinical datasets with distinct temporal patterns. Top row (A–D): Kaplan-Meier curves with HCHG-flagged intervals (shaded). Bottom row (E–H): per-interval *signed* $-\log_{10}$(hypergeometric p-value) — upward bars mark excess events in the treatment arm, downward bars excess in the control arm. Dashed lines are the per-direction HCHG thresholds; bars reaching beyond either line are highlighted in red and coincide exactly with the intervals shaded above. **(A,E)** CheckMate 057 PFS (nivolumab vs. docetaxel, $n = 582$); log-rank $p = 0.351$, HCHG $p = 0.002$. Curves cross at ∼3 months before nivolumab gains a durable late advantage. **(B,F)** COMET-1 OS (cabozantinib vs. prednisone, $n = 1028$); log-rank $p = 0.262$, HCHG $p = 0.012$. Early OS advantage (months 3–7) during bone-lesion response, fading as resistance develops. **(C,G)** AZURE DFS (zoledronic acid vs. control, $n = 3359$); log-rank $p = 0.305$, HCHG $p = 0.012$. Menopause-dependent benefit accumulates in months 20–60. **(D,H)** CSL OS (prednisone vs. placebo, liver cirrhosis, $n = 446$); log-rank $p = 0.384$, HCHG $p = 0.002$. Time-varying corticosteroid effect with multiple curve crossings; *fully public* raw IPD (CRAN `timereg` / SurvSet). In all four datasets the log-rank test and four weighted alternatives are non-significant. For the first three, IPD reconstructed via Guyot et al. [2012].

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